

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The Problem–Solution Fit Canvas helps identify the target users, understand their challenges, and evaluate how the proposed solution addresses those challenges effectively. For the **Smart Lender – Loan Eligibility Prediction System**, it highlights the problems faced by loan applicants and bank credit officers and demonstrates how an AI-powered loan prediction system provides faster, accurate, and data-driven decision-making.

1 CUSTOMER SEGMENTS  Loan Applicants Individuals seeking personal, home, education, or business loans. Bank Credit Officers Bank staff responsible for evaluating loan applications and making credit decisions.	6 CUSTOMER LIMITATIONS  For Loan Applicants - Limited financial literacy. - Mostly mobile users with limited access to detailed financial tools.  For Bank Credit Officers - Limited time to evaluate each application. - Rely on basic desktops and legacy systems with limited analytics.	5 AVAILABLE SOLUTIONS 1. Manual Underwriting  Pros: Human oversight, personal touch.  Cons: Slow process, Human bias, Costly operations 2. Legacy Rule-Based Software  Pros: Faster decisioning compared to manual process.  Cons: Rigid rules, Cannot learn from historical data, Inaccurate for self-employed applicants
2 PROBLEMS / PAINS + FREQUENCY  - Slow loan approval process - Human bias in decision-making - Human error in evaluation - Weeks of waiting for applicants	9 PROBLEM ROOT / CAUSE  Lack of automation and data-driven decisioning. - Manual cross-referencing of 10+ financial metrics including: - Income - Dependents - Credit history - Property area - And other key financial factors  Result: Time-consuming process, prone to human bias and error, leading to delays and inconsistent decisions.	7 BEHAVIOR + INTENSITY  - Loan applicants frequently call or check status due to long waiting periods. - Credit officers spend ~80% of their day verifying documents and cross-checking data. - Constant pressure to meet turnaround times with limited tools and high workloads.  INTENSITY: HIGH (Causes stress, delays, and operational inefficiency)
3 TRIGGERS TO ACT [TR]  - Applicants need urgent financial assistance and submit loan applications. - Banks receive a large backlog of loan requests daily. - Need for faster, accurate, and unbiased loan decisions.	10 YOUR SOLUTION Smart Lender – Loan Eligibility Prediction System (Flask Web Application)  - Powered by XGBoost ML model for accurate predictions. - Captures 11 financial metrics from applicants. - Automatically normalizes and processes data. - Sleek Glassmorphism UI for modern user experience. - Provides real-time loan decision and risk band (Low / Medium / High). - Eliminates manual review bottleneck and reduces decision time from weeks to seconds.  Outcome: Faster, unbiased, consistent, and data-driven loan eligibility decisions.	8 CHANNELS OF BEHAVIOR  1. Online - Bank web portals for applications and status tracking. - Email notifications and follow-ups.  2. Offline - Branch visits, long queues. - Customer-service phone calls for status and clarifications.
4 EMOTIONS BEFORE / AFTER  BEFORE (Applicants) Anxious, impatient, uncertain, overwhelmed due to long waiting time and unclear status.  AFTER (Applicants) Relieved, confident, and informed due to instant, unbiased feedback and clear loan decision.		